

Homework — 1.3.4 Web Technologies — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. (a) [1] An element/tag selector — selecting on the tag name `p`, which applies the style to all `<p>` elements (1).
- (b) [2] An `#id` selector (1). Justification (1): the warning box is a **single** element and an id must be unique to one element on the page, whereas a class is intended for a style shared by several elements. (The HTML already gives the element `id="warning"`.)
2. [2] Any two for 1 mark each: the styling is defined once and applied to every page, giving a consistent appearance; one change to the CSS file updates the whole site, making maintenance far easier; it separates content (HTML) from presentation (CSS); the HTML files are smaller/cleaner and the CSS file can be cached by the browser. Max 2.
3. [2] A web crawler/spider automatically visits web pages and follows the hyperlinks it finds to discover further pages (1). The words/content and metadata it collects are used by the search engine to build an index mapping keywords to the pages they occur on, so results can be retrieved quickly (1).
4. (a) [2] Each incoming link acts as a "vote" for the importance of the page it points to (1). A link from a high-ranking page passes on more rank / carries more weight than a link from a low-ranking page, so it increases the target page's rank by more (1).
- (b) [2] Iteratively: every page's rank is repeatedly recalculated using the most recent rank values of the pages that link to it (1). Converges: the repetition stops when the rank values settle / no longer change significantly between iterations (1).
5. (a) [1] Any one: the user gets instant feedback without waiting for a round trip to the server; it reduces load on the server / network traffic, as no request is sent for a simple check (1).
- (b) [2] Only the server has access to the database of existing usernames — the browser cannot see that data (1). Also, client-side code can be viewed, modified or bypassed by the user, so a client-side check cannot be trusted; the check must run on the server where it cannot be tampered with (1).

Part B (6 marks)

6. [2] A switch connects devices *within* a LAN and forwards frames to the intended device using **MAC** addresses (1). A router connects *different* networks (e.g. a LAN to the internet) and forwards packets between them using **IP** addresses (1).
7. [2] Symmetric encryption uses the *same* secret key to encrypt and decrypt, so the key itself must be transferred to the other party — and if it is intercepted in transit, all security is lost (1). Asymmetric encryption uses a public/private key pair: the public key can be shared openly because only the never-shared private key can decrypt, so no secret ever needs to be exchanged (1).

8. [2] A primary key is an attribute that uniquely identifies each record in a table (1). A foreign key is an attribute in one table (e.g. `CustomerID` in `Order`) that holds/references the primary key of another table, linking each order to its customer (1).

Total: 14 + 6 = 20 marks